Style Before Memory — Experiment Plan

Oct 19, 2026

We first confirm the core claim with a small pilot (G0 and G1, 17 LoRA runs), then scale up only if it holds. Two dates decide everything: **W4 (G1: go or no-go for ICML)** and **W6 (G2: fix d_style and d_memory)**.

Overview

Main claim to test. When a VLM is fine-tuned on benchmark items, it picks up the benchmark's caption *style* from far less exposure than it needs to *memorize* the reference captions. We want to show which part of the score gain is style, which is item memory, and whether contamination detectors can tell them apart.

RQ	Question	Experiments
RQ1	Does style adaptation appear at a lower exposure than reference memorization?	G1, G2, G3, C2
RQ2	Which part of the exposure (image, caption, pairing, breadth vs. depth, reference variety) causes what?	C1, C2, C3, C4
RQ3	Is the score gain real quality, or inflation from style?	B2, B3, B5, G4
RQ4	Do detectors catch exposure, memorization, or inflation?	B6, C9
Generality	Other models, realistic contamination, update scope, a second task	C5 to C8, C10, C11

Three tracks.

- **Track A, gates (G0 to G4).** Run in order. Each has a pass rule and a fallback written down before we see results.
- **Track B, independent work (B1 to B8).** Needs no exposed checkpoints, so it starts in W1. B3 and B5 give our first RQ3 results with zero training.
- **Track C, extensions (C1 to C11).** Start only after G1 or G2 passes.

Ground rules.

1. Pass rules and margins are fixed in `prereg.md` before results come in. Thresholds are chosen on split A and confirmed on a fresh split B.
2. Every checkpoint is scored on the same four parts: **S** (style), **A** (item association), **V** (verbatim memory), **Q** (real quality).
3. All conditions use the same exposed items. Smaller exposure sets are subsets of larger ones.
4. The d=0 baseline is shared by every split, so it is trained once.
5. RQ3 and RQ4 need no new training. Keep every checkpoint and every generation (T=0 and T=1).

Decisions to lock in W1

These seven go into `prereg.md` before any exposed run starts. D2 is the one that sets the size of the whole plan, so it is decided together with a short timing run in W1.

ID	Decision	Proposal	Owner
D1	Main model	Keep LLaVA-1.5-7B if the B1 audit passes, otherwise SmolVLM-2.2B. LLaVA's stage-2 data (665K) is public, which C5 needs.	Kaustav
D2	Budget per run	Fix total examples M for every run. Exposed examples K = N_exp × d replace general examples. Set M and N_exp from a W1 timing run (see below).	Minho
D3	Caption and Image condition formats	Caption: fixed text-only template. Image: the base model's own caption as the target, **not** a length-matched text (length is our main style signal).	Kaustav
D4	Same-distribution control	COCO train captions (no overlap with eval images), matched to NoCaps in count, length and **total K**.	Kaustav
D5	Decision margins	δ_S, δ_A = 2 × SD across five d=0 seeds. δ_V = 10% of the Paired d=64 V effect, with an absolute floor. δ_Q = 5 points of human win rate.	Yoorhim
D6	Replication model	SmolVLM-2.2B or Molmo-7B-O, whichever is not the main model. Training data must be public so we can audit NoCaps exposure.	Kaustav
D7	Seeds	Separate training, split and reference-assignment seeds. Split A = explore, split B = confirm.	Minho

Why M matters (D2). NoCaps val has 4,500 images. Exposing 50% gives N_exp = 2,250, so K is 2,250 / 9,000 / 36,000 / 144,000 at d = 1 / 4 / 16 / 64. To keep d=64 under half the data, M must be about 300k, and 96+ runs of 300k examples on a 7B model may exceed our GPU budget. If the W1 timing run says so, we start with N_exp = 1,000 and M = 130k instead (same shares, less than half the cost).

M (total per run)	d=1 share	d=4	d=16	d=64
300k, N_exp = 2,250	0.75%	3%	12%	48%
130k, N_exp = 1,000	0.8%	3.1%	12%	49%

Shared metrics

Every checkpoint gets the same metric set, reported as four parts. **Bold** = the primary metric used for gate decisions.

Part	Metrics	Measured on
S, style	**Change in caption length and boilerplate rate on unseen images**, sentence count, function-word and POS mix, style classifier (B4)	Unseen images (seen also logged)
A, item association	**Δ_d,c (eq. 3, 9 held-out refs)**, Γ_d (eq. 4)	Seen minus unseen
V, verbatim memory	**ΔRMI: RMI minus the d=0 model's RMI on the same image**, rank of the training ref among 10, exact match, longest shared n-gram	Seen images, teacher-forced
Q, real quality	CIDEr, SPICE, **CIDEr on style-counterfactual refs (B2)**, CLIPScore, LLM judge, human pairwise	All images
Detection	DCQ, CDD (T=1, 20 samples per image, 50 if unstable), KDS	All images

$$\mathrm{RMI}_i = \log p(r_i^{train} \mid x_i) - \frac{1}{9}\sum_{j=1}^{9} \log p(r_{ij}^{held} \mid x_i)$$

- Generate at T=0 by default. Sample again at T=1 from the same checkpoint for CDD and robustness.
- Report image-level paired bootstrap CIs, with per-seed and pooled values.
- Threshold τ = the smallest d where the effect's CI lower bound exceeds δ, from a monotone fit over $\log_2 d$.

How the experiments run

All gate runs use the main model, LoRA on the language model plus projector (vision encoder frozen), and standard dilution.

Track A: gates

The gates take 46 new runs (52 with the optional d=8). G4 reuses G2 checkpoints and needs no training.

ID	Question	Conditions	New runs	Pass rule (pre-registered)	If it fails
G0	Is the pipeline right, and how noisy is it?	Split A, d=0 × 5 seeds + untrained base	5	Base CIDEr matches reported range. Seed SD sets δ_S, δ_A. GPU hours per run confirmed.	Hold G1 until the eval pipeline is fixed
G1	Can we see style and memory effects separately? (pilot)	Split A, {Caption, Paired, SameDist} × d ∈ {4, 64} × 2 seeds	12	(a) Paired d=64 ΔRMI CI above 0 and exact match clearly above d=0, which sets δ_V. (b) Unseen S change at d=4 exceeds δ_S. (c) The P0 grid fits our compute.	(a) Stop and fix exposure or LoRA. (b) Try d=16; if still nothing, drop “style first” and pivot to “decomposing contamination gains + detectors”. (c) Apply the cut list; if still short, move to NeurIPS 2027.
G2	Where are τ_style and τ_memory? (RQ1 main)	Caption, Paired at d ∈ {1, 16} × 3 seeds, third seed for d ∈ {4, 64}, SameDist d=16 × 3. Add d=8 if the gap is wide.	21 (+6)	**Primary:** at d_style, the upper 95% CI of the V effect is below δ_V (equivalence test). Secondary: CI of τ_memory − τ_style above 0 on $\log_2$ d. Fix d_style and d_memory here.	If both effects start at the same d, reframe as “dose-response per component” and skip C3
G3	Does the order replicate on a new split?	Split B, {Caption, Paired} × {d_style, d_memory} × 2 seeds, d=0 shared	8	Same order on split B, effect sizes inside split A’s CI	Report the failure, add seeds, limit the claim to split A
G4	Is the low-exposure gain real quality or style inflation? (RQ3 main)	G2 checkpoints at d ∈ {0, d_style, d_memory} × {Caption, Paired, SameDist}: counterfactual refs, output neutralizing, reference-free metrics, length control, LLM judge, blind human pairwise on 2 key contrasts	0	Call it “inflation” only if all three hold: original-ref score up, human and grounded quality gain below δ_Q, and the gain shrinks under neutral refs and neutralized outputs	If quality rises too, reframe as “harmless style adaptation that CIDEr cannot tell apart from contamination”. RQ4 still stands.

- If SameDist shows the same S change as Caption in G1, that is a finding, not a failure: low-exposure gains look like ordinary style fine-tuning, which strengthens RQ3 and RQ4.

Track B: independent work, starting W1

ID	Work	Output	Used by	Done by
B1	Training data audit of main and replication models: NoCaps val / Open Images use, near-duplicate image search	Audit memo per model, D1 and D6 decided	All runs	W1
B2	Style-counterfactual references: 300 seen + 300 unseen images, 2–3 held-out captions each rewritten in original, neutral and verbose styles (LLM rewrite, human spot-check that objects and relations are kept)	Checked 3-style reference set	G4, B4, C3, C9	W4
B3	Prompt intervention: base and d=0 models with neutral, NoCaps-like (“one short sentence”) and verbose prompts	How much CIDEr and SPICE move from style alone	G4 baseline, early paper finding	W2
B4	Style metric suite + content-controlled style classifier (trained on B2 rewrites)	S metrics, classifier accuracy	S in every gate	W4 (G1 uses length and boilerplate first)
B5	Metric sensitivity: score one held-out human ref against the others, before and after style rewrites and length changes	Model-free curve of how much metrics reward style	Intro motivation, G4	W3
B6	Detectors: DCQ, CDD, KDS. Check behavior and variance on base and d=0, run on G1 checkpoints as soon as they exist	Detector code, d=0 reference distribution	C9	W4
B7	Port the replication model: same prompts and eval, reproduce its d=0 baseline, measure compute per run	Replication-model G0 results	C8	W5
B8	Related work (incl. Bordt 2025, Hubble, Spiesberger 2026, Jayaraman 2024), ICML template, impact statement	Related Work draft	Paper	W3

Track C: extensions after the gates

C1 and C2 need no thresholds, so they start right after G1, in parallel with G2. The rest run only at the d_style and d_memory points fixed in G2. **C2 is the core evidence for RQ1** (see the note under the table). P0 = answers a direct reviewer objection, P1 = mechanism detail, P2 = appendix or stretch.

ID	Question	Conditions	New runs	After	Priority
C1	Which of image, caption or pairing drives S and V (Γ_d)?	{Image, Shuffled} × d ∈ {1, 4, 16, 64} × 3 seeds. Shuffled pairs only across classes or low CLIP similarity.	24	G1	P0
C2	Does breadth drive style and depth drive memory?	Same K = 2,250: (B, d) = (50%, 1), (12.5%, 4), (3.125%, 16) × {Caption, Paired} × 3 seeds. Nested sets, seen metrics on the shared 140 images. (50%, 1) reused from G2.	12	G1	**P0, core**
C2b	Does C2 hold at K = 9,000?	(12.5%, 16), (3.125%, 64) × {Caption, Paired} × 3 seeds	12	C2	P1
C3	Can exact wording be separated from meaning?	Paired at {d_style, d_memory}, refs {multi-ref, paraphrase, counterstyle} × 2 seeds, all scored on the same 5 held-out refs	12	G2, B2	P1
C4	Does style survive 10× dilution?	{Caption, Paired} × d_style × 10× M × 2 seeds	4 (≈ 40 runs of compute)	G2	P1
C5	Realistic contamination: same pattern inside the full instruction mix?	Insert into LLaVA-665K: {Caption, Paired} × {d_style, d_memory} + one clean 665K run, 1 seed	5 (large)	G2	P0
C6	Sensitive to update scope or near-duplicates?	Paired × {d_style, d_memory} broader fine-tuning × 2 seeds + Paired d_memory crop/re-encode × 2 seeds	6	G2	P2
C7	Does style spread to unseen classes?	Class-disjoint split, {Caption, Paired} × d_style × 2 seeds	4	G2	P2
C8	Does the order replicate on a second model?	Replication model, {Caption, Paired} × d ∈ {0, 4, 16, 64} plus d_style and d_memory if off-grid × 2 seeds, SameDist d_style × 2, rerun G4 interventions	16+	G2, B7	P0
C9	What do detectors actually catch?	No training. Label every checkpoint by exposure and behavior, then DCQ/CDD AUROC and PR-AUC per regime, KDS correlation with S, V, inflation, false positives on d=0, SameDist and counterstyle. Also test **counterfactual-ref scoring as a style-corrected metric**.	0	G4, C1, C3, C8, B6	P0
C10	Second task: on short-answer VQA or multiple choice (e.g. MMStar), does “answer format” come first?	Paired × {d_style, d_memory} × 2 seeds (small version)	4	G2, task design	P1
C11	Third model (Molmo-7B-O)	2–3 key contrasts only	≈ 6	C8	P2

- **Why C2 is core.** In the main grid, raising d also raises K, the total number of NoCaps-style captions. Style likely tracks K and memory tracks d, so “style comes first along d” is partly built into the design. C2 holds K fixed and trades breadth for depth, which is what actually separates the two. Our headline claim should be “style follows K, memory follows d”.
- C3’s counterstyle checkpoints double as C9’s counterstyle control.
- C2’s 3.125% arm is only about 140 images, so seen-group power is low. If pilot variance is too high, treat C2b (K = 9,000) as the main result.

Run budget

Priority	Experiments	New training runs	What must not be dropped
P0	G0 to G3, C1, C2, C8 + C5 (large) + G4, C9 (no training)	98 + 5 large	Every main-paper figure
P1	C2b, C3, C4 (≈ 40 runs of compute), C10 small, G2 optional d=8	38 (≈ 74 runs of compute)	Mechanism detail, dilution, second task
P2	C6, C7, C11	16	Appendix, stretch

If compute runs short, cut in this order. G0 to G4, C2, C5, C8 and C9 stay to the end.

1. Drop C11
2. C6 and C7 down to 1 seed
3. C4 Paired only (2 runs)
4. C1 to d ∈ {4, 16, 64} × 2 seeds (24 to 12)
5. C3 multi-ref and counterstyle only (12 to 8)
6. Drop C2b
7. Reduce N_exp to 1,000 and M to 130k (if not already done in W1)

Week-by-week timeline

W1 is the week of Oct 19, so W14 is the week of Jan 18, in line with a late-January ICML deadline (2027 dates not yet announced).

Week	Track A (gates)	Track B (independent)	Track C (extensions)	Decision
W1 (Oct 19)	Timing run, lock D1–D7, start G0 training	Start B1, B3, B8	–	Main model, M and N_exp
W2 (Oct 26)	G0 eval	Finish B3; B2, B4, B5, B6 in progress	–	δ_S, δ_A
W3 (Nov 2)	G1 training	Finish B5, B8	–	–
W4 (Nov 9)	G1 eval	Finish B2, B4, B6	–	**G1: go or no-go for ICML**
W5 (Nov 16)	G2 training	B7	Start C2	–
W6 (Nov 23)	G2 eval	Finish B7	C2 continues, start C1	**G2: fix d_style, d_memory**
W7 (Nov 30)	G3 training, G4 automatic metrics	–	C1 continues; start C3, C4, C5, C8	–
W8 (Dec 7)	G4 human eval	–	C2b, C5, C8, C10	–
W9 (Dec 14)	G3 eval	–	C6, C7 training, start C9	**G3, G4 verdicts**
W10 (Dec 21)	Class final report	–	C8 G4 interventions, C9 continues	–
W11 (Dec 28)	–	–	Finish C9, P2 if time allows	Main figures locked
W12–W14 (Jan 4–22)	Paper writing	–	Patch gaps reviewers would hit	Submit

What each member does

Each list is in week order. Tick items off as you go.

Minho: training and running experiments

Owns every training run and the go or no-go calls.

- ☐ W1: Timing run (about 1k steps) to measure GPU hours per run, then fix M and N_exp (D2)
- ☐ W1: Splits and seed scheme (D7); start G0 training (d=0, 5 seeds)
- ☐ W3: Train G1 pilot (12 runs)
- ☐ W4: Make the ICML go or no-go call with the team (G1)
- ☐ W5–W6: Train G2 (21 runs) and C2 (12 runs); start C1 (24 runs)
- ☐ W7–W8: Train G3, C3, C4, C5 (665K runs), C8 (replication model), C2b, C10
- ☐ W9: Train C6, C7 if compute allows
- ☐ W12–W14: Paper intro, method and setup sections; merge the full draft

Yoorhim: evaluation and analysis

Owns the metrics, the margins and the gate verdicts.

- ☐ W1: Set decision margins (D5) and write `prereg.md` with the team
- ☐ W1–W2: Eval pipeline for S, A, V (incl. ΔRMI), Q; run B3 prompt intervention with Kaustav’s prompts
- ☐ W2: Evaluate G0, fix δ_S and δ_A
- ☐ W2–W4: Build B4 style metrics and the style classifier
- ☐ W4: Evaluate G1, report pass or fail for each criterion
- ☐ W6: Evaluate G2, fix d_style and d_memory
- ☐ W7: G4 automatic metrics (counterfactual refs, reference-free, length control, LLM judge)
- ☐ W5–W10: Evaluate every new checkpoint as it lands; bootstrap CIs and monotone fits
- ☐ W9: Evaluate G3
- ☐ W12–W14: Results section and figures, appendix

Kaustav: VLM judgment and detectors

Owns model choice, condition formats and everything about detectors.

- ☐ W1: B1 training data audit, then choose the main and replication models (D1, D6)
- ☐ W1: Lock condition formats (D3) and the SameDist control (D4); write the B3 prompts
- ☐ W2–W4: Implement DCQ, CDD, KDS (B6) and run them on base, d=0 and G1
- ☐ W3–W4: Spot-check Injoh's B2 rewrites
- ☐ W5–W6: Port the replication model (B7) and reproduce its d=0 baseline
- ☐ W6–W7: Design the second task (C10)
- ☐ W8–W10: Evaluate C8 on the replication model
- ☐ W9–W11: C9 detector analysis, incl. the style-corrected score
- ☐ W12–W14: Detector and qualitative analysis sections

Injoh: references, metric checks and paper support

Owns the B2 reference set, B5 and the paper's housekeeping.

- ☐ W1–W3: Related work (B8), incl. Bordt 2025, Hubble, Spiesberger 2026, Jayaraman 2024
- ☐ W2–W3: Metric sensitivity analysis (B5), no model needed
- ☐ W2–W4: Make B2 style-counterfactual references (LLM rewrites of 600 images)
- ☐ W3: ICML template, impact statement, check reciprocal-reviewer eligibility
- ☐ W8: Run the G4 blind human pairwise eval (2 key contrasts), hiding condition labels
- ☐ W12–W14: Finish Related Work, references, anonymization check